

```
from google.colab import files
uploaded = files.upload()
image_path = list(uploaded.keys())[0]

import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread(image_path)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(6, 6))
plt.imshow(gray, cmap='gray')
plt.title("Original Grayscale Image")
plt.axis('off')
plt.show()
```

Choose Files images (2).jpg

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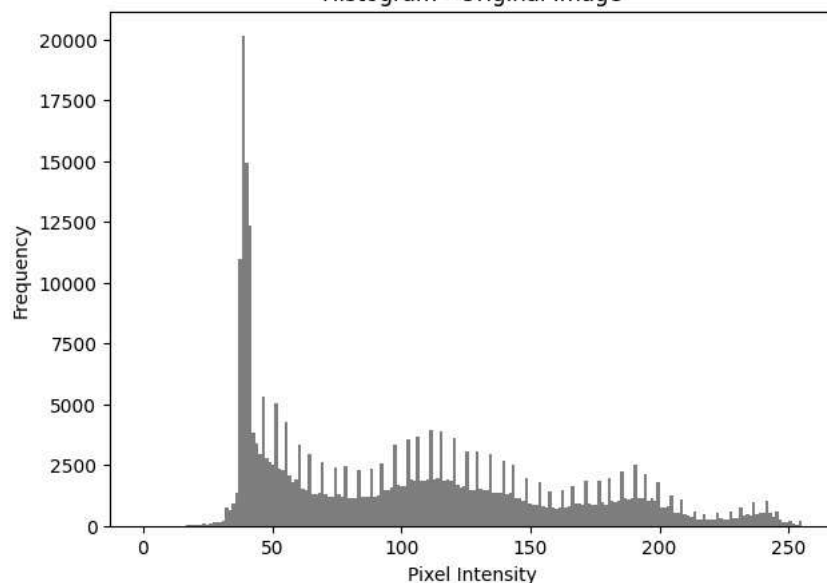
Original Grayscale Image



```
import cv2
import numpy as np
import matplotlib.pyplot as plt

plt.figure(figsize=(7, 5))
plt.hist(gray.ravel(), bins=200, range=[0, 255], color='gray')
plt.title("Histogram - Original Image")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.show()
```

Histogram - Original Image



```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
def contrast_stretch(img):
    min_val = np.min(img)
    max_val = np.max(img)
    stretched = (img - min_val) * (255.0 / (max_val - min_val))
    return stretched.astype(np.uint8)

stretched_img = contrast_stretch(gray)

plt.figure(figsize=(6, 6))
plt.imshow(stretched_img, cmap="gray")
plt.title("Contrast Stretched Image")
plt.axis('off')
plt.show()
```

Contrast Stretched Image



```
import cv2
import numpy as np
import matplotlib.pyplot as plt

equalized_img = cv2.equalizeHist(gray)

plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original")
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(equalized_img, cmap="gray")
plt.title("Histogram Equalized")
plt.axis('off')

plt.show()
```

Original



Histogram Equalized



```
import cv2
import numpy as np
import matplotlib.pyplot as plt

clahe = cv2.createCLAHE(clipLimit=2.5, tileGridSize=(10, 10))
clahe_img = clahe.apply(gray)

plt.figure(figsize=(6, 6))
```

```
plt.imshow(clahe_img, cmap='gray')
plt.title("CLAHE Enhanced Image")
plt.axis('off')
```

```
plt.show()
```

CLAHE Enhanced Image



```
import cv2
import numpy as np
import matplotlib.pyplot as plt

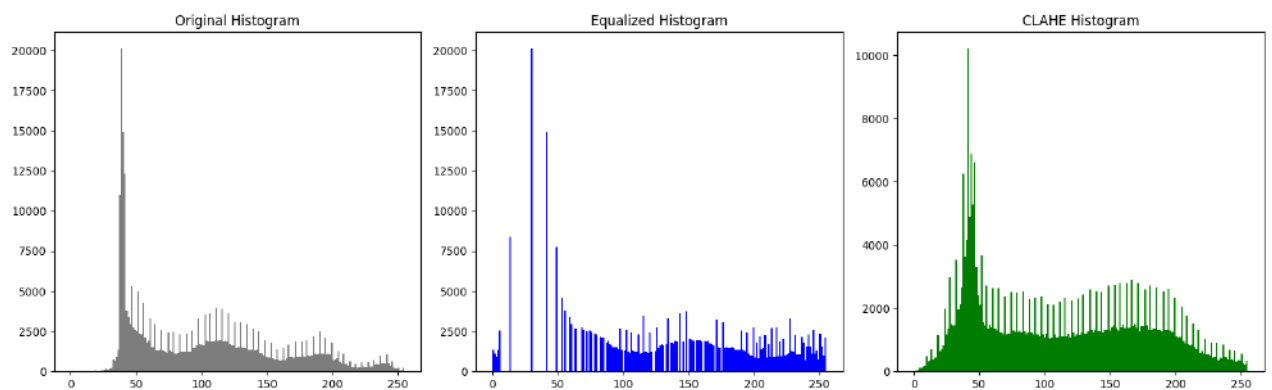
fig, axs = plt.subplots(1, 3, figsize=(16, 5))

axs[0].hist(gray.ravel(), bins=200, range=[0, 255], color='gray')
axs[0].set_title("Original Histogram")

axs[1].hist(equalized_img.ravel(), bins=200, range=[0, 255], color="blue")
axs[1].set_title("Equalized Histogram")

axs[2].hist(clahe_img.ravel(), bins=200, range=[0, 255], color="green")
axs[2].set_title("CLAHE Histogram")

plt.tight_layout()
plt.show()
```



```
import cv2
import numpy as np
import matplotlib.pyplot as plt

fig, axs = plt.subplots(1, 4, figsize=(20, 5))

axs[0].imshow(gray, cmap="gray")
axs[0].set_title("Original")
axs[0].axis('off')

axs[1].imshow(stretched_img, cmap="gray")
axs[1].set_title("Contrast Stretched")
```

```
axs[1].axis('off')  
  
axs[2].imshow(equalized_img, cmap="gray")  
axs[2].set_title("Histogram Equalized")  
axs[2].axis('off')  
  
axs[3].imshow(clahe_img, cmap='gray')  
axs[3].set_title("CLAHE")  
axs[3].axis('off')  
  
plt.tight_layout()  
plt.show()
```

